Feature spec: project presets for VerdeCity Planner

Context

The VerdeCity Planner tool allows urban planners in Porto Alegre to create tree-planting scenarios where they calculate the cost and thermal impact of planting trees on different streets. Currently, users configure everything manually. This feature adds **predefined presets** that suggest an optimal configuration based on the selected street type, which the user can then freely edit.

Expected behavior

User flow

1. The user selects a street or group of streets on the map (or picks a neighborhood).
2. The system detects the **street type** (arterial with median, arterial without median, collector, open local, dense local, BRT corridor) using the road classification from OSM or the city's Master Plan (Plano Diretor).
3. The user is shown the **recommended preset** for that street type, with a banner like "Suggested configuration for arterial road with median".
4. The preset pre-fills all scenario fields: species, quantities, spacing, planting side, etc.
5. The user can **accept the preset as-is** and view the calculations, or **edit any field** (change species, adjust spacing, modify mix percentages, etc.).
6. When the user edits a value, the badge changes from "Suggested preset" to "Custom (based on Arterial with median)" so they know they've modified the recommendation.
7. Cost and impact calculations are recalculated in real time with each change.

How the preset is assigned

Assignment is based on street type. If the street has a road classification attribute, it's used directly. Otherwise, it's inferred from width:

Street width	Assigned type
⩾ 25m and has a median/central divider	Arterial with median
20-30m without median	Arterial without median
15-20m	Collector
12-15m with buildings ⩽ 3 stories average	Open local
10-14m with buildings > 4 stories average	Dense local
Street tagged as a transit corridor	BRT corridor

If the system cannot determine the type, show a manual selector with the 6 types for the user to choose.

Data for the 6 presets

Preset 1: Arterial with median

- **Display name**: "Arterial green corridor"
- **Short description**: "Avenues with central median (⩾25m total width)"
- **Urban canyon type**: Open (H/W < 0.5)
- **Example streets in Porto Alegre**: Av. Assis Brasil, Av. Ipiranga, Av. Protásio Alves
- **Typical width**: 25-40m total
- **Trees per block (100m)**: 26
- **Tree spacing**: 7m
- **Target canopy coverage**: 45%
- **Estimated air temperature reduction (at maturity)**: 2.0 – 2.5 °C
- **Estimated surface temperature reduction (at maturity)**: 5 – 8 °C
- **Time to full maturity**: 8-12 years

Suggested species mix:

Species	Common name	% of mix	Size	Leaf type	Location on block
Tipuana tipu	Tipuana	30%	Large (15-25m)	Semi-deciduous	Median (central divider)
Handroanthus avellanedae	Ipê-roxo	25%	Large (15-25m)	Deciduous	Median + north sidewalk
Peltophorum dubium	Canafístula	20%	Large (15-25m)	Semi-deciduous	North sidewalk
Vitex megapotamica	Tarumã	15%	Large (10-20m)	Evergreen	South sidewalk
Eugenia uniflora	Pitangueira	10%	Medium (5-10m)	Evergreen	Interspersed between large trees

Placement instructions: Median: large trees every 10m. North sidewalk: large trees every 8m. South sidewalk: medium-size every 10m. Maintain gaps between canopies aligned with the prevailing E-SE wind direction.

Estimated costs per 100m block (in R$):

Component	Cost
Seedlings (26 trees)	R$ 2,080
Planting + infrastructure	R$ 7,800
Maintenance year 1-2	R$ 3,120
Maintenance year 3-5	R$ 6,240
Maintenance year 6-10 (estimated)	R$ 3,744
Total 10 years per block	**R$ 22,984**
Cost per tree (10 years)	R$ 884

User-facing notes: "Maximum thermal impact. The median allows large trees with no overhead cable conflicts. Prioritize the north side for pedestrian sidewalk shade. Keep canopy gaps in the median for E-SE wind corridors. Ideal for public transit corridors."

Preset 2: Arterial without median

- **Display name**: "Lateral arterial boulevard"
- **Short description**: "Wide avenues without central median (20-30m)"
- **Urban canyon type**: Medium-open (H/W 0.5-1.0)
- **Examples**: Av. Bento Gonçalves, Av. Osvaldo Aranha, R. Voluntários da Pátria
- **Typical width**: 20-30m
- **Trees per block (100m)**: 16
- **Spacing**: 8m
- **Target canopy coverage**: 35%
- **Air temp. reduction**: 1.5 – 2.0 °C
- **Surface temp. reduction**: 3.5 – 6 °C
- **Maturity**: 8-12 years

Species mix:

Species	Common name	%	Size	Leaf	Location
Handroanthus avellanedae	Ipê-roxo	30%	Large	Deciduous	North sidewalk
Peltophorum dubium	Canafístula	25%	Large	Semi-deciduous	North sidewalk
Cordia americana	Guajuvira	20%	Large	Dense canopy	South sidewalk
Andira fraxinifolia	Angelim-doce	15%	Medium (6-12m)	Evergreen	Under power lines
Psidium cattleyanum	Araçazeiro	10%	Medium (3-8m)	Evergreen	Interspersed

Placement: North sidewalk: large trees every 8m (priority). South sidewalk: medium/large every 10m. Alternate deciduous and evergreen for year-round shade.

Costs per 100m block:

Component	Cost
Seedlings (16 trees)	R$ 1,280
Planting + infrastructure	R$ 4,800
Maintenance year 1-2	R$ 1,920
Maintenance year 3-5	R$ 3,840
Maintenance year 6-10	R$ 2,304
Total 10 years	**R$ 14,144**
Per tree	R$ 884

Notes: "Without a median, planting is concentrated on sidewalks. North side is priority (more sun in the southern hemisphere). Check for overhead power lines: where cables exist, use only medium-size trees (Angelim-doce, Araçazeiro). Leave gaps every 3 trees aligned with the prevailing E-SE wind."

Preset 3: Collector

- **Display name**: "Green collector street"
- **Short description**: "Neighborhood collector roads (15-20m wide)"
- **Canyon type**: Medium (H/W 0.7-1.5)
- **Examples**: R. Gen. Lima e Silva, R. Cel. Bordini, R. Dr. Barcelos
- **Width**: 15-20m
- **Trees / 100m**: 12
- **Spacing**: 9m
- **Target canopy**: 30%
- **Air temp. reduction**: 1.2 – 1.8 °C
- **Surface temp. reduction**: 3 – 5 °C
- **Maturity**: 7-10 years

Species mix:

Species	Common name	%	Size	Leaf	Location
Handroanthus chrysotrichus	Ipê-amarelo	30%	Medium-large (8-15m)	Deciduous	North side
Vitex megapotamica	Tarumã	25%	Large (10-20m)	Evergreen	North side
Eugenia involucrata	Cerejeira-do-mato	20%	Medium (up to 12m)	Evergreen	Both sides
Andira fraxinifolia	Angelim-doce	15%	Medium (6-12m)	Evergreen	Under power lines
Eugenia uniflora	Pitangueira	10%	Medium (5-10m)	Evergreen	Interspersed

Placement: North side only as priority. South side if budget allows. 9m spacing to avoid closing the canyon. Clear trunks above 3m for pedestrian flow and vehicle visibility.

Costs per 100m block:

Component	Cost
Seedlings (12 trees)	R$ 840
Planting + infrastructure	R$ 3,600
Maintenance year 1-2	R$ 1,440
Maintenance year 3-5	R$ 2,880
Maintenance year 6-10	R$ 1,728
Total 10 years	**R$ 10,488**
Per tree	R$ 874

Notes: "Balance between cooling and urban compatibility. Medium urban canyon requires attention to ventilation — don't fully close the canopy. Prioritize tall clean trunks (>3m) to avoid obstructing pedestrian flow or vehicle visibility."

Preset 4: Open local

- **Display name**: "Open residential street"
- **Short description**: "Local roads in low/medium density neighborhoods (12-15m)"
- **Canyon type**: Low (H/W < 0.5, 1-3 story houses)

- **Examples**: Residential streets in Sarandi, Rubem Berta, Restinga
- **Width**: 12-15m
- **Trees / 100m**: 10
- **Spacing**: 10m
- **Target canopy**: 30%
- **Air temp. reduction**: 1.5 – 2.0 °C
- **Surface temp. reduction**: 4 – 7 °C
- **Maturity**: 8-12 years

Species mix:

Species	Common name	%	Size	Leaf	Location
Enterolobium contortisiliquum	Timbaúva	25%	Large (20-30m) umbrella canopy	Deciduous	Sidewalk with root space
Handroanthus avellanedae	Ipê-roxo	25%	Large (15-25m)	Deciduous	Alternating with Timbaúva
Eugenia uniflora	Pitangueira	20%	Medium (5-10m)	Evergreen	Tight spaces
Psidium cattleyanum	Araçazeiro	15%	Medium (3-8m)	Evergreen	Interspersed
Calliandra brevipes	Topete-de-cardeal	15%	Shrub (1-3m)	Evergreen	Base planters around trees

Placement: Both sides when space allows. Prioritize blocks with no existing trees. Supplement with native shrubs at the base for layered vegetation.

Costs per 100m block:

Component	Cost
Seedlings (10 trees + shrubs)	R$ 600
Planting + infrastructure	R$ 2,500
Maintenance year 1-2	R$ 1,000
Maintenance year 3-5	R$ 2,000
Maintenance year 6-10	R$ 1,200
Total 10 years	**R$ 7,300**
Per tree	R$ 730

Notes: "Maximum potential in vulnerable neighborhoods (Sarandi, Rubem Berta). Without tall buildings, wide canopy works perfectly. Include low shrub vegetation as a complement. The Timbaúva generates spectacular umbrella-type shade but needs ample root space. These neighborhoods have the highest social impact and should be prioritized."

Preset 5: Dense local

- **Display name**: "Dense residential street"
- **Short description**: "Narrow local roads with medium/tall buildings (10-14m)"
- **Canyon type**: Deep (H/W > 1.5, 6+ story buildings)
- **Examples**: Streets in Centro Histórico, Bom Fim, Cidade Baixa, Menino Deus
- **Width**: 10-14m
- **Trees / 100m**: 8
- **Spacing**: 12m
- **Target canopy**: 20%
- **Air temp. reduction**: 0.8 – 1.2 °C
- **Surface temp. reduction**: 2 – 4 °C
- **Maturity**: 6-10 years

Species mix:

Species	Common name	%	Size	Leaf	Location
Handroanthus chrysotrichus	Ipê-amarelo	30%	Medium (8-15m) tall trunk	Deciduous	Sun-exposed side
Andira fraxinifolia	Angelim-doce	30%	Medium (6-12m) globular canopy	Evergreen	Alternating
Eugenia involucrata	Cerejeira-do-mato	20%	Medium (up to 12m) compact canopy	Evergreen	Tight spaces
Eugenia uniflora	Pitangueira	20%	Medium (5-10m)	Evergreen	Under power lines

Placement: Only the side receiving the most direct sun (varies by street orientation). Tall clean trunk (>4m) for pedestrian-level ventilation. Do not overlap canopies — leave gaps of at least 3m between canopies.

Costs per 100m block:

Component	Cost
Seedlings (8 trees)	R$ 480
Planting + infrastructure	R$ 2,000
Maintenance year 1-2	R$ 800
Maintenance year 3-5	R$ 1,600
Maintenance year 6-10	R$ 960
Total 10 years	**R$ 5,840**
Per tree	R$ 730

Notes: "IMPORTANT: Deep canyon — DO NOT use dense canopies or close the canopy. This traps heat at pedestrian level at night. Use open-canopy species with tall clean trunks (>4m). Wider spacing (12m) to allow ventilation. Buildings already provide partial shade, so the tree's role here is to supplement with evapotranspiration rather than shade. Consider complementing with green facades and green roofs where possible."

Preset 6: BRT corridor

- **Display name**: "Mass transit corridor"
- **Short description**: "Bus corridors with exclusive lanes (≥30m)"
- **Canyon type**: Variable (depends on segment)
- **Examples**: Av. Assis Brasil (corridor), Av. Bento Gonçalves (T1-T2 segments)
- **Width**: 30-40m+
- **Trees / 100m**: 20
- **Spacing**: 8m
- **Target canopy**: 35%
- **Air temp. reduction**: 1.8 – 2.5 °C
- **Surface temp. reduction**: 4 – 7 °C
- **Maturity**: 8-12 years

Species mix:

Species	Common name	%	Size	Leaf	Location
Peltophorum dubium	Canafístula	30%	Large (15-25m)	Semi-deciduous	Lateral separators along the corridor
Handroanthus avellanedae	Ipê-roxo	25%	Large (15-25m)	Deciduous	Pedestrian sidewalks
Vitex megapotamica	Tarumã	20%	Large (10-20m)	Evergreen	Adjacent to bus stops
Eugenia involucrata	Cerejeira-do-mato	15%	Medium (up to 12m)	Evergreen	Interspersed
Andira fraxinifolia	Angelim-doce	10%	Medium (6-12m)	Evergreen	Under power lines

Placement: Lateral separators of the corridor + sidewalks. Evergreens (Tarumã) always adjacent to bus stops for year-round shade. Deciduous trees in intermediate stretches. Maintain a minimum 5m distance from traffic signs and traffic lights.

Costs per 100m block:

Component	Cost
Seedlings (20 trees)	R$ 1,600
Planting + infrastructure	R$ 6,000
Maintenance year 1-2	R$ 2,400
Maintenance year 3-5	R$ 4,800
Maintenance year 6-10	R$ 2,880
Total 10 years	**R$ 17,680**
Per tree	R$ 884

Notes: "Prioritize shade at bus stops and access sidewalks. People waiting for the bus are among the most exposed to heat. Use evergreen trees (Tarumã) at every stop to guarantee year-round shade including winter. Maintain minimum distance from traffic signs and traffic lights (5m). Co-benefit: reduces temperature inside parked and idling buses."

UI implementation

Preset component structure

```
PresetSelector
├── StreetTypeDetector (auto-detects or lets user pick manually)
├── PresetCard (shows the recommended preset)
|   ├── PresetHeader (name, canyon badge, description)
|   ├── StatsGrid (trees/100m, spacing, canopy, thermal impact)
|   ├── SpeciesMixEditor (editable table with % per species)
|   ├── PlacementGuide (placement text)
|   ├── CostBreakdown (recalculable cost table)
|   └── NotesPanel (contextual recommendations)
└── EditModeToggle (switch between "suggested preset" and "custom")
```

User-editable fields

After loading a preset, the user should be able to edit:

1. **Tree spacing** — slider from 6m to 15m, in 1m steps. When changed, the number of trees per block is automatically recalculated (= block length / spacing × sides planted).
2. **Species mix** — the user can:
    - Change the percentage of each species (percentages must sum to 100).

    - Remove a species from the mix.
    - Add a new species from the general catalog.
    - Replace one species with another.
3. **Planting sides** — toggle: "North side only", "South side only", "Both sides", "Median + both sides". This affects the tree quantity multiplier.
4. **Block length** — numeric input, defaults to 100m. If the user selected a real street from the map, auto-fill with the actual segment length.
5. **Number of blocks** — multiplier to scale the project. Slider or input from 1 to 100 blocks.
6. **Unit costs** — if the reference costs don't apply (e.g., a nursery has different prices), the user can edit unit costs for seedlings, planting, and maintenance.

Real-time recalculation

Every time the user changes a value, automatically recalculate:

- **Total tree count** = (block length / spacing) × sides × number of blocks. Round to the nearest integer.
- **Count per species** = total × percentage of each species. Round.
- **Estimated canopy coverage** = $\Sigma(\text{species_count} \times \pi \times (\text{canopy_diameter}/2)^2) / (\text{block_length} \times \text{street_width}) \times 100$. Canopy diameter comes from the species catalog.
- **Air temperature reduction** = Δcanopy(%) × 0.08 × canyon_factor × orientation_factor. Factors are derived from the preset's canyon type.
- **Total cost** = Σ(species_count × seedling_cost_per_species) + (total_trees × planting_cost) + projected maintenance costs.
- **Cost per °C of reduction** = total_cost / ΔTair (cost-effectiveness metric).

Thermal impact calculation factors

These factors adjust the base coefficient of 0.08 °C per 1% increase in canopy coverage:

Factor	Value	When it applies
Kclimate (Porto Alegre, Cfa)	0.08	Always (base)
Kcanyon open (H/W < 0.5)	1.2	Arterial with median, Open local
Kcanyon medium (H/W 0.5-1.5)	1.0	Arterial without median, Collector, BRT
Kcanyon deep (H/W > 1.5)	0.7	Dense local
Korientation E-W	1.0	East-west oriented streets
Korientation N-S	0.85	North-south oriented streets
Kcontiguous network (≥10 adjacent blocks)	1.15	If the user selects 10+ contiguous blocks
Kdispersed (isolated blocks)	0.85	If blocks are not contiguous
Kpavement (trees over asphalt)	1.3	If the street is paved (default yes)

Formula: $\Delta Tair = \Delta canopy(\%) \times Kclimate \times Kcanyon \times Korientation \times Kcontiguity \times Kpavement$

Always display a **±30% uncertainty band** on temperature estimates.

Species catalog for the selector

When the user wants to add or replace a species, they are shown a filtered dropdown. Each species in the catalog has these attributes (which also feed the calculations):

Attribute	Type	Example
scientific_name	string	"Handroanthus avellanedae"
common_name	string	"Ipê-roxo"
origin	enum: native_rs, native_br, exotic	native_rs
size_class	enum: small, medium, large	large
mature_height_min	number (m)	15
mature_height_max	number (m)	25
canopy_diameter_min	number (m)	8
canopy_diameter_max	number (m)	12
leaf_type	enum: evergreen, deciduous, semi_deciduous	deciduous
lai	number	4.2
allergenic_potential	enum: low, medium, high	low
power_line_compatible	boolean	false
drought_tolerance	enum: low, medium, high	medium
root_aggressiveness	enum: low, medium, high	medium
branch_hazard_risk	enum: low, medium, high	low
problematic_fruits	boolean	false
seedling_cost_min	number (R$)	50
seedling_cost_max	number (R$)	150
available_municipal_nursery	boolean	true
years_to_maturity	number	10
cooling_coefficient	number (°C per standard tree)	0.15

The dropdown should be filterable by: size class (compatible with street width), power line compatibility (if the street has overhead cables), and origin (prioritize native RS species per the Master Plan requirement of 70% regional natives).

Initial species catalog data

Include at least these species at launch:

Large size:

1. Handroanthus avellanedae (Ipê-roxo) — deciduous, 15-25m, canopy 8-12m
2. Peltophorum dubium (Canafístula) — semi-deciduous, 15-25m, canopy 8-12m
3. Vitex megapotamica (Tarumã) — evergreen, 10-20m, canopy 8-12m
4. Enterolobium contortisiliquum (Timbaúva) — deciduous, 20-30m, canopy 12-15m
5. Cordia americana (Guajuvira) — evergreen, 10-20m, canopy 6-10m
6. Tipuana tipu (Tipuana) — semi-deciduous, 15-25m, canopy 10-15m
7. Parapiptadenia rigida (Angico-vermelho) — deciduous, 15-25m, canopy 6-10m
8. Cedrela fissilis (Cedro) — deciduous, 15-25m, canopy 6-10m

Medium size: 9. Handroanthus chrysotrichus (Ipê-amarelo) — deciduous, 8-15m, canopy 4-6m 10. Andira fraxinifolia (Angelim-doce) — evergreen, 6-12m, canopy 4-6m 11. Eugenia involucrata (Cerejeira-do-mato) — evergreen, 6-12m, canopy 4-6m 12. Eugenia uniflora (Pitangueira) — evergreen, 5-10m, canopy 3-5m 13. Psidium cattleyanum (Araçazeiro) — evergreen, 3-8m, canopy 3-5m 14. Allophylus edulis (Chal-chal) — evergreen, 6-10m, canopy 3-5m 15. Myrcianthes pungens (Guabiju) — evergreen, 8-15m, canopy 4-6m

Shrub (complement): 16. Calliandra brevipes (Topete-de-cardeal) — evergreen, 1-3m 17. Schinus terebinthifolius (Aroeira-vermelha) — evergreen, 5-10m, canopy 4-6m

Impact visualization

When the user has a configured scenario (either preset or edited), display:

1. **Thermal impact bar**: horizontal colored bar showing the temperature reduction range. Bar color gradient from red (low impact) to green (high impact). Show the range with ±30% uncertainty.
2. **"Before vs. after" comparison**: two large numbers side by side — "Current estimated temperature: 47.3°C (surface)" → "Post-intervention temperature: ~40°C". Use LST data from the selected neighborhood as the baseline.
3. **Maturation timeline**: simple line chart showing the projected temperature reduction from year 0 to year 20, with the sigmoidal maturation curve (30% at year 3, 70% at year 7, 95% at year 12).
4. **Cost-effectiveness summary**: card with "R$ X per °C of reduction per block" — allows quick comparison between presets.

"Reset to preset" behavior

Add a "Restore original preset" button that reverts all edited values to the preset defaults.

Confirm with a modal: "Restore suggested configuration? Unsaved changes will be lost."

Implementation notes

- All costs are reference values from March 2026 for Porto Alegre. They must be editable by a system administrator.
- Thermal impact coefficients (0.08 °C/%, adjustment factors) come from a synthesis of peer-reviewed scientific literature (meta-analysis of 115+ studies). They are planning estimates with ±30% uncertainty.
- Porto Alegre's Urban Tree Planting Master Plan (Plano Diretor de Arborização) requires 70% regional native species. The system should validate that the user's mix meets this requirement and show a warning if it doesn't.
- Species flagged as "invasive exotic" in the Instituto Hórus database must NOT appear in the catalog.